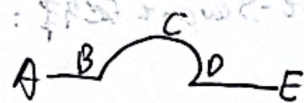


9.4 圆心O处的B是由AB、DE和 $\frac{1}{2}$ 圆弧共同叠加而成的.

BCD在O处 $\vec{B}_1 = \frac{\mu_0}{4\pi} \int \frac{Id\vec{l}}{r^2} = \frac{\mu_0}{4\pi} \int_0^{2\pi} \frac{I r d\theta}{r^2} = \frac{\mu_0 I}{6r}$. 垂直纸面向里



AB导线产生的 $\vec{B}_2$ 大小: $B_2 = \frac{\mu_0 I}{4\pi a} (\cos\theta_1 - \cos\theta_2)$, $\theta_1 = 0$, $\theta_2 = \frac{5\pi}{6}$

其中 $a = r \cos 60^\circ = \frac{1}{2}r$ $\therefore B_2 = \frac{\mu_0 I}{2\pi r} (1 - \frac{\sqrt{3}}{2})$ 垂直纸面向里

DE产生的 $B_3 = \frac{\mu_0 I}{4\pi a} (\cos\theta'_1 - \cos\theta'_2)$, $\theta'_1 = \frac{5\pi}{6}$, $\theta'_2 = \pi$

$\Rightarrow B_3 = \frac{\mu_0 I}{2\pi r} (1 - \frac{\sqrt{3}}{2})$ 也向里

$\Rightarrow B = B_1 + B_2 + B_3 = \frac{\mu_0 I}{r} (\frac{7}{6} - \frac{\sqrt{3}}{2\pi})$

9.7 以O为原点, 沿AB向下为x轴正方向, AB上点到O点距离为 $[a, a+b]$

线段以 ω 绕O旋转时, 距O点为x, 长度dx微元, 带电量为 $dq = \lambda dx$.

又 $T = \frac{2\pi}{\omega}$, $\therefore$ 等效为环形电流, $dI = \frac{dq}{T} = \frac{\lambda\omega}{2\pi} dx$.

由毕奥萨伐尔知, $dB = \frac{\mu_0 dI}{2x} = \frac{\mu_0 \lambda \omega}{4\pi x} dx$

$B = \int_a^{a+b} \frac{\mu_0 \lambda \omega}{4\pi x} dx = \frac{\mu_0 \lambda \omega}{4\pi} \ln(1 + \frac{b}{a})$, 方向垂直旋转平面向上

2. $dm = dI \cdot S = (\frac{\lambda\omega}{2\pi} dx)(\pi x^2) = \frac{1}{2} \lambda \omega x^2 dx$ 对 $[a, a+b]$ 积分: $m = \int_a^{a+b} \frac{1}{2} \lambda \omega x^2 dx = \frac{1}{6} \lambda \omega [(a+b)^3 - a^3]$
 $= \frac{1}{6} \lambda \omega [3a^2b + 3ab^2 + b^3]$
 $= \frac{\lambda \omega b}{6} [3a^2 + 3ab + b^2]$

9.11 $S_1 = \pi r_1^2$, $S_2 = \pi(r_3^2 - r_2^2)$. 由对称性, 距轴线为 r 的地方, 磁感应方向沿同心圆切线方向. 取半径为 r 同心圆作为安培环路, $\oint B \cdot dl = B \cdot 2\pi r$

区域一: $0 \leq r < r_1$ 时, $I_{\text{内}} = I \cdot \frac{\pi r^2}{\pi r_1^2} = I \frac{r^2}{r_1^2}$

$$\Rightarrow B \cdot 2\pi r = \mu_0 I \frac{r^2}{r_1^2} \quad \therefore B = \frac{\mu_0 I r}{2\pi r_1^2}$$

二: $r_1 \leq r < r_2$ 时, 环路完全包含内圆柱, 未接触到外圆筒.

$I_{\text{内}} = I$, 由安培环路定理

$$B \cdot 2\pi r = \mu_0 I \Rightarrow B = \frac{\mu_0 I}{2\pi r}$$

三: $r_2 \leq r < r_3$, $I_{\text{内}} = I - I \frac{\pi(r^2 - r_2^2)}{\pi(r_3^2 - r_2^2)} = I \frac{r_3^2 - r^2}{r_3^2 - r_2^2}$

由环路定理:

$$B \cdot 2\pi r = \mu_0 I \frac{r_3^2 - r^2}{r_3^2 - r_2^2} \Rightarrow B = \frac{\mu_0 I}{2\pi r} \frac{r_3^2 - r^2}{r_3^2 - r_2^2}$$

四: $r \geq r_3$, 去路、来路电流大小相等, 方向相反.

$$I_{\text{内}} = I - I = 0 \Rightarrow B \cdot 2\pi r = 0 \Rightarrow B = 0$$

$$\therefore B = \begin{cases} \frac{\mu_0 I r}{2\pi r_1^2}, & 0 \leq r < r_1, \\ \frac{\mu_0 I}{2\pi r}, & r_1 \leq r < r_2, \\ \frac{\mu_0 I (r_3^2 - r^2)}{2\pi r (r_3^2 - r_2^2)}, & r_2 \leq r < r_3, \\ 0, & r \geq r_3 \end{cases}$$

9.14 圆柱形导体可看作电流沿Y轴正向、半径为 R_1 的圆柱导体
与沿Y轴负向、半径 R_2 的圆柱导体组成。

$$I_1 = \frac{I}{\pi(R_1^2 - R_2^2)} \cdot \pi R_1^2 = \frac{R_1^2}{R_1^2 - R_2^2} I$$

$$I_2 = \frac{I}{\pi(R_1^2 - R_2^2)} \cdot \pi R_2^2 = \frac{R_2^2}{R_1^2 - R_2^2} I$$

(1) 由叠加原理, 轴线上点O的 $B_0 = B_{R_1} + B_{R_2}$

又 $R_1 = 0 \Rightarrow B_{R_1} = 0$, 由安培环路定理得

$$B_0 = B_{R_2} = \frac{\mu_0 I R_2^2}{2\pi a (R_1^2 - R_2^2)} = 2 \times 10^{-6} T$$

(2) 空腔轴线: $B_{01} = B_{01}' + B_{02}'$, $R_2 = 0, B_{02}' = 0$

$$\Rightarrow B_{01}' = \frac{\mu_0 I a}{2\pi (R_1^2 - R_2^2)} = 2 \times 10^{-4} T$$

(3)

9.15 由左手定则知方向向右。

$$F_A = G = mg, F_A = BIL$$

$$\therefore I = \frac{mg}{BL} = 0.4 A$$

9.16 由安培力公式, $d\vec{F} = I d\vec{l} \times \vec{B}$

$$\vec{F} = \int_L d\vec{F} = \int_L I d\vec{l} \times \vec{B}$$

又由对称性可知, 合力向上, $F_{\text{合}} = \int_L I d\vec{l} \times \vec{B}$

$$F_{\text{合}} = \int_L dF \sin \theta = \int_L I dl \cdot B \sin \theta = IB \sin \theta \int_L dl$$

垂直于圆环平面向下。 $= 2\pi l B \sin \theta$

9.30. $r = \frac{l}{v}$, Y方向洛伦兹力

$$\text{洛伦兹力提供向心力 } qvB = \frac{mv^2}{R} \Rightarrow R = \frac{mv}{qB}$$

$$\text{由几何关系, } \sin \theta = \frac{l}{R} = \frac{qBl}{mv}$$

又由最终总偏转角Y与整个大三角形: $\tan \theta = \frac{y}{L}$

$$\Rightarrow y = L \tan \theta = L \frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}} = L \frac{\frac{qBl}{mv}}{\sqrt{1 - (\frac{qBl}{mv})^2}} \approx \frac{qBl^2}{mv}$$

9.32 由动能定理 $Vq = \frac{1}{2}mv^2$.

由洛伦兹力提供向心力 $qvB = \frac{mv^2}{R}$

其中 $r = \frac{x}{2}$.

$$\text{代入 } v = \sqrt{\frac{2Vq}{m}}, \Rightarrow qB = \frac{m}{r} \sqrt{\frac{2Vq}{m}}$$

$$\text{平方 } q^2 B^2 = \frac{2mVq}{(\frac{x}{2})^2}$$

$$\Rightarrow M = \frac{B^2 q}{8V} x^2$$

补充二. $\vec{B} = \frac{\mu_0 I_2}{2\pi\rho} \vec{e}_\varphi, \rho = \sqrt{x^2 + y^2}$.

电流向纸外 $\Rightarrow \vec{B} = \frac{\mu_0 I_2}{2\pi(x^2 + y^2)} (-y\vec{i} + x\vec{j})$.

圆上点 $\begin{cases} x = R\cos\theta, \\ y = R(1 + \sin\theta) \end{cases} \Rightarrow d\vec{l} = R(-\sin\theta\vec{i} + \cos\theta\vec{j})d\theta$.

$$\text{又 } x^2 + y^2 = 2R^2(1 + \sin\theta), \vec{B} = \frac{\mu_0 I_2}{4\pi R} \left(-\vec{i} + \frac{\cos\theta}{1 + \sin\theta} \vec{j} \right)$$

1. 微元受力: $d\vec{F} = I_1 d\vec{l} \times \vec{B}$

$$\text{代入: } d\vec{F}_z = \frac{\mu_0 I_1 I_2}{4\pi} \left(-\cos\theta - \sin\theta \frac{\cos\theta}{1 + \sin\theta} \right) d\theta$$

$$\Rightarrow F_z = -\frac{\mu_0 I_1 I_2}{4\pi} \left[\ln(1 + \sin\theta) \right]_0^{2\pi} = 0 \therefore F = 0$$

对O'圆心取力矩: $d\vec{M} = \vec{r}' \times d\vec{F}, \vec{r}' = R(\cos\theta\vec{i} + \sin\theta\vec{j}), d\vec{F} = dF_z \vec{k}$.

$$\Rightarrow d\vec{M} = -\frac{\mu_0 I_1 I_2}{4\pi} \frac{\cos\theta}{1 + \sin\theta} (\sin\theta\vec{i} - \cos\theta\vec{j}) d\theta, M_x = \int_0^{2\pi} \frac{\sin\theta \cos\theta}{1 + \sin\theta} d\theta \left(-\frac{\mu_0 I_1 I_2 R}{4\pi} \right), \mu_0 \text{ 为真空磁导率.}$$

补充一. 设平板在 xOy 平面, 电流沿 $+x$ 方向流动.

由对称性知, 磁场必平行于平板且垂直于电流方向.
 $z > 0$, 磁场指 $-y$; $(\frac{d}{2})$ 外. $(\frac{1}{2}By)$

$z < 0$, 磁场指 $+y$; 仅与 $|z|$ 相关. $(\frac{d}{2})$

$$\int_{-d/2}^{d/2} \vec{H} \cdot d\vec{l} = I$$

设导体上下紧邻磁场分别为

$$H_y(z^+) = H_0, H_y(z^-) = -H_0,$$

$$\Rightarrow H_0 - (-H_0) = I \Rightarrow H_0 = \frac{I}{2}$$

$$\Rightarrow H = \begin{cases} \frac{I}{2}, & z > 0 \\ -\frac{I}{2}, & z < 0 \end{cases}$$

而在 $z = \pm \frac{d}{2}$ 处无自由面电流.

$$\vec{n} \times (\vec{H}_2 - \vec{H}_1) = 0$$

$$\therefore H_y(\frac{d^-}{2}) = H_y(\frac{d^+}{2}) = \frac{I}{2} \therefore \text{整个回, } H \text{ 不变.}$$

$$\text{ii) } |z| < \frac{d}{2} \text{ 时, } B = \begin{cases} \frac{\mu_0 I}{2}, & 0 < z < \frac{d}{2} \\ -\frac{\mu_0 I}{2}, & -\frac{d}{2} < z < 0 \end{cases}$$

$$\text{真空中 } (|z| > \frac{d}{2}) \quad B = \begin{cases} \frac{\mu_0 I}{2}, & z > \frac{d}{2} \\ -\frac{\mu_0 I}{2}, & z < -\frac{d}{2} \end{cases}$$

$$\text{得 } \vec{M}_x = 0, \quad \vec{M}_y = \frac{\mu_0 I_1 I_2 R}{4\pi} \int_0^{2\pi} \frac{\cos^2 \theta}{1 + \sin \theta} d\theta = \frac{\mu_0 I_1 I_2 R}{4\pi} \int_0^{2\pi} (1 - \sin \theta) d\theta = \frac{1}{2} \mu_0 I_1 I_2 R \vec{e}_y.$$

$$\therefore \vec{M} = \vec{M}_y = \frac{1}{2} \mu_0 I_1 I_2 R \vec{e}_y.$$

由右手定则, 电流逆时针. $\vec{m} = I_1 \pi R^2 \vec{k}$ (指向纸外)

$\vec{M} = +\vec{e}_y$, 又由 $\frac{d\vec{I}}{dt} = \vec{M}$ 知, 线圈绕 y 轴转
从 $+y$ 看, 顺时针

(3) 圆周上每个点到导线距离均为 R . $B = \frac{\mu_0 I_2}{2\pi R}$ 恒定. 沿切向.
又 $d\vec{I}$ 也沿切向 $\Rightarrow d\vec{I} \parallel \vec{B}$.

$$\therefore d\vec{F} = I_1 d\vec{I} \times \vec{B} = 0. \quad \therefore \vec{M} = 0.$$